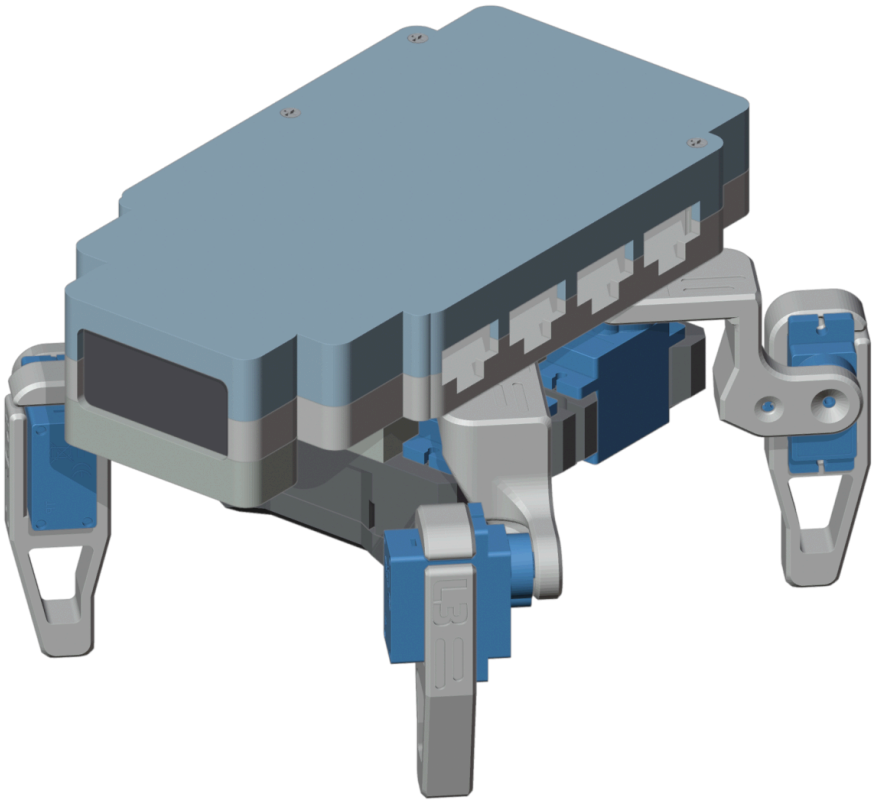
Atech-crab

Assembly guide · print files v4 (wire openings on both sides) · 3 November 2026



Atech-crab is Sesame, Dorian Todd's open-source quadruped (Apache-2.0), with an Atech board as its brain. Its frame, servos and legs stay as Sesame makes them, and a new deck and lid from Atech Atelier ("Sesame walker" 0.1.3) carry an Atech 14-port board flat on top, with the screen in a visor at the front. The eight servos plug into the board's ports 2–5 on jumper wires.

New in v4

- **Slots on both sides:** at servo ports 2–5 and at ports 10–13, half in the lid and half in the deck, so the jumpers go straight out of the headers.
- **SG90 lead notches** under each slot, down to the gap under the board, on both sides.
- **USB-C opening** reaches 5 mm lower, through the seam: part of it is cut from the lid, part from the deck.

> **Not built yet.** These pictures come from the CAD, which passes Atelier's fit check. Nobody has printed and assembled v4. Test-fit the deck on the frame, with the servos in, before you print a second set.

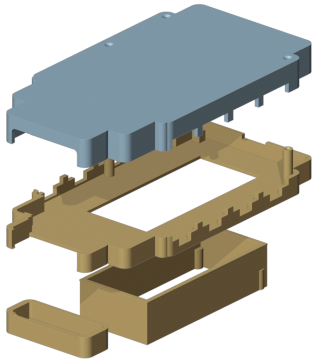
What you need

Printed (13 parts, see atech/print/README.md for your printer's file)

Part	Qty	From
Internal frame	1	Sesame
Bottom cover	1	Sesame
Hip legs (femurs) L1, L2, R1, R2	4	Sesame
Feet L3, L4, R3, R4	4	Sesame
Deck part 1 (plate)	1	Atelier, v4
Deck part 2 (spine and head)	1	Atelier
Lid	1	Atelier, v4

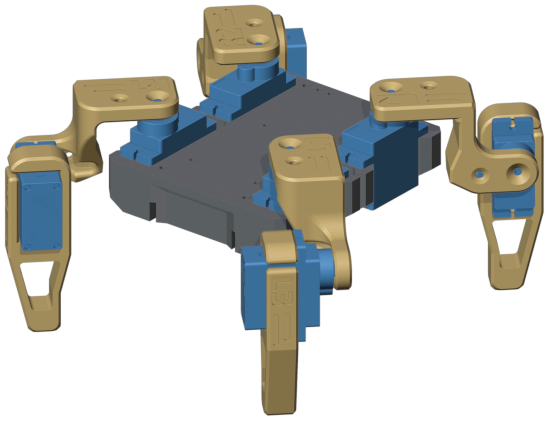
Hardware

Item	Qty
SG90 or MG90S servos	8
Atech 14-port board	1
Atech USB-C module (port 8)	1
Atech motion sensor module, icm40608 (port 13)	1
Atech screen module and its cable (ports 10+11)	1
M2×8 pan-head screws (deck 4, bottom cover 2)	6
M2×10 countersunk screws (lid)	3
Male-to-male jumper wires	12+
5 V 3 A supply for the servos	1
Superglue (CA)	—



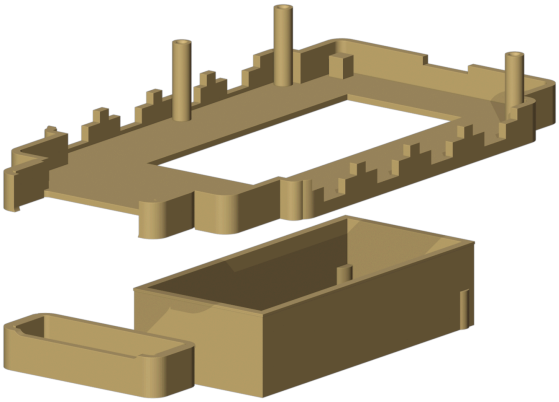
The three Atelier parts: lid (top), deck part 1, deck part 2. In the step pictures, the parts you add in that step are in colour; the parts already in place are pale grey; printed parts you add are amber.

1 Build the base



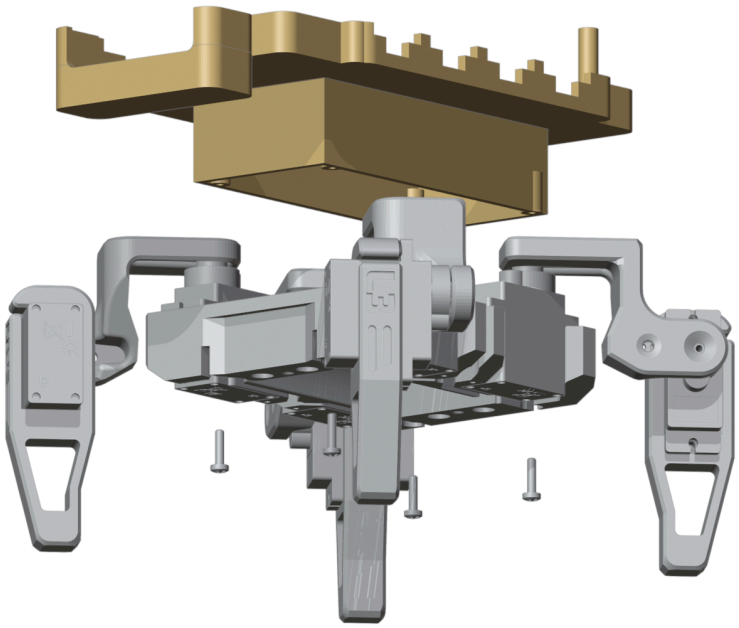
1. Put the four hip servos into the frame, and the four knee servos into the femurs, as in Sesame's own guide (github.com/dorianborian/sesame-robot).
2. Fit the femurs and the feet onto the servo horns.
3. Run every servo lead into the frame, up towards its top opening.
4. Leave the bottom cover off for now: the deck screws go in from inside the frame.

2 Glue the deck



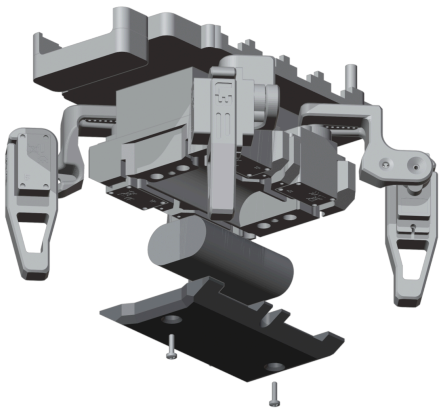
1. Dry-fit deck part 2 (spine and head) under deck part 1. The 0.6 mm lip fits the recess, so it only goes one way.
2. Glue it with superglue (CA) and hold it until it sets.

3 Deck onto the frame



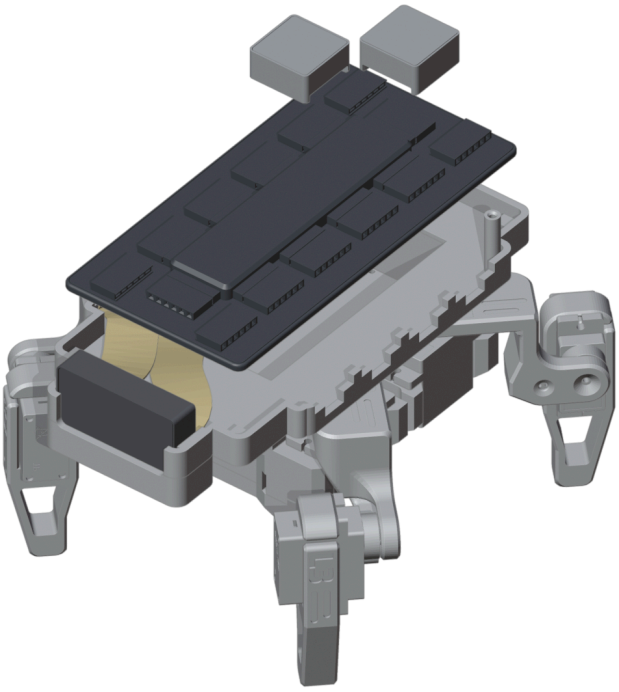
1. Feed all eight servo leads up through the deck's spine as you lower the deck onto the frame.
2. From inside the frame, through its open bottom, drive 4 × M2×8 pan-head screws up into the deck's bosses.

4 Battery and bottom cover



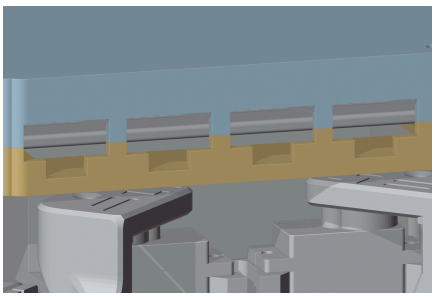
1. Put Sesame's battery into the frame.
2. Close the bottom with the bottom cover: 2 × M2×8 pan-head screws.

5 Board, modules and screen

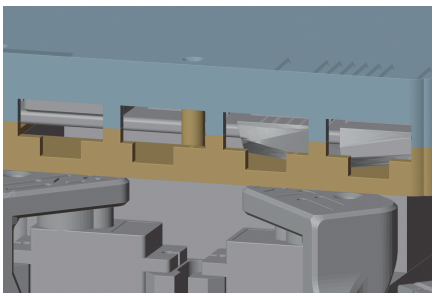


1. Pull the servo leads out to the side of ports 2–5 (or out of the notches on the other side, if that is shorter); they lie in the 5 mm gap under the board.
2. Set the board on the four posts in the tray, ports 1–6 on the side the leads come out, port 8 at the back.
3. Slide the USB-C module into port 8 (its socket faces the back wall), and the motion sensor into port 13.
4. Stand the screen upright in the head, behind the window, connector edge down. Its two ribbons loop down into the right-hand bay, run flat under the board and plug into ports 10 and 11.

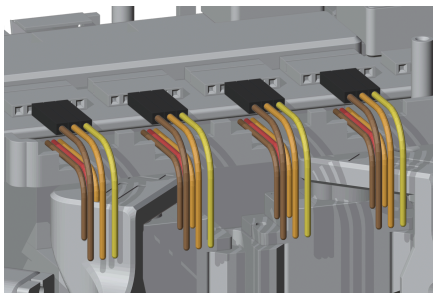
6 Wire the servos



The slots at ports 2–5, half lid, half deck, and the lead notches under them.



The same on the other side, at ports 10–13. Port 12's notch sits 2 mm back, clear of a lid-screw post.



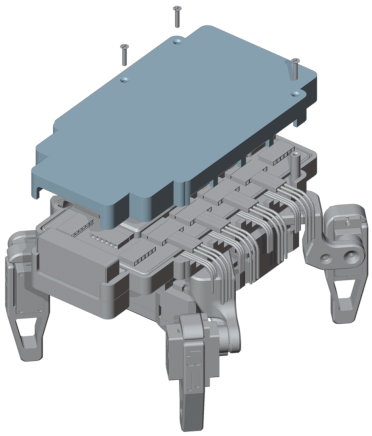
Jumpers out of the headers, servo leads out of the notches (an illustration, not pin-accurate).

Port	Leg	Line A (pin 5)	Line B (pin 6)
2	front left	L1 hip	L3 knee
3	front right	R1 hip	R3 knee
4	back left	L2 hip	L4 knee
5	back right	R2 hip	R4 knee

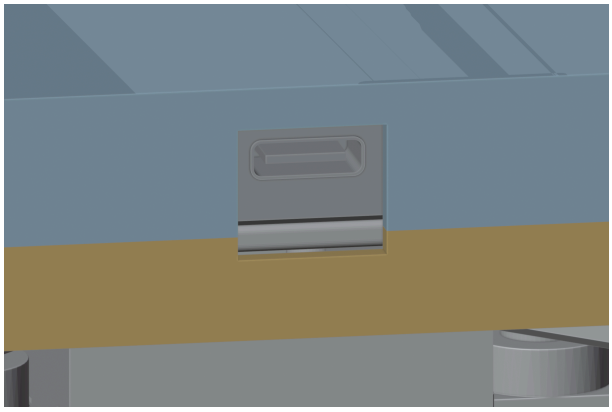
1. Per port, three jumpers: 🟤 GND (pin 4), 🟠 line A (pin 5) to the hip servo's signal, 🟡 line B (pin 6) to the knee servo's signal. Plug them in straight, through the slot.
2. Bring each servo lead out through the notch under its port and join it to its jumper outside the wall: the servo's orange (or yellow) wire to the signal jumper.
3. Servo power: every servo's red wire to the 5 V supply's +, every brown wire to its –, and the supply's – to a GND jumper.

> **+12 V sits next to +5 V.** Port pins: 1 +12 V, 2 +5 V, 3 +3.3 V, 4 GND, 5 line A, 6 line B. Which end of the header is pin 1 is not marked. Measure with the board on USB: the pin at 3.3 V is pin 3, its neighbour that beeps to the USB-C shell is GND (pin 4), and lines A and B follow. Never connect pin 1.

7 Close the lid



1. Check that no ribbon or lead is pinched between the deck and the lid, and that each jumper sits in its slot.
2. Fit the lid and drive 3 × M2×10 countersunk screws into the bosses in the side bays.



The USB-C opening at the back, now 13 × 11.5 mm, across the seam.

8 First run

1. Plug a USB-C data cable into the back, and switch on the servos' 5 V supply.
2. Flash the walker: in `atech/firmware`, run `atech upload .`
3. Check the board: `atech check`. Then try one move with `atech send spider_go wave`, holding the robot off the table.

The walker firmware and its gaits have been run only in simulation (an emulated board in MuJoCo), not yet on a built robot. A servo that only twitches is short of power: use the separate 5 V supply.

